

$$N1. \int \frac{x^2 - x + 1}{\sqrt{x}} dx = \int x^{3/2} dx - \int x^{1/2} dx + \int x^{-1/2} dx = \frac{x^{5/2} \cdot 2}{5} - \frac{2x^{3/2}}{3} + \frac{2x^{1/2}}{1} + C$$

$$N2. \int \frac{dx}{7+x^2} = \frac{1}{\sqrt{7}} \arctan \frac{x}{\sqrt{7}} + C$$

$$N3. \int \frac{dx}{3x^2-5} = \frac{1}{3} \int \frac{dx}{x^2 - \frac{5}{3}} = -\frac{1}{3} \int \frac{dx}{\left(\frac{\sqrt{5}}{3}\right)^2 - x^2} = -\frac{1}{3} \cdot \frac{1 \cdot \sqrt{3}}{2 \cdot \sqrt{5}} \ln \left| \frac{\frac{\sqrt{5}}{3} + x}{\frac{\sqrt{5}}{3} - x} \right| + C$$

$\left(\frac{\sqrt{5}}{3}\right)^2 = a^2$

$$N4. \int \sin^2 \frac{x}{2} dx = \int \frac{1 - \cos x}{2} dx = \int \frac{dx}{2} - \frac{1}{2} \int \cos x dx = \frac{1}{2} x - \frac{1}{2} \sin x + C$$

$$N5. \int \tan^2 x dx = \int \frac{\sin^2 x}{\cos^2 x} dx = \int \frac{1 - \cos^2 x}{\cos^2 x} dx = \int \frac{dx}{\cos^2 x} - \int dx = \tan x - x + C$$

$$N6. \int e^{3x} dx = \frac{1}{3} \int e^{3x} d(3x) = \frac{1}{3} \int e^t dt = \frac{1}{3} e^t + C = \frac{1}{3} e^{3x} + C$$

$dx = \frac{1}{3} d(3x)$

$$N7. \int \sin(2x+3) dx = \frac{1}{2} \int \sin(2x+3) d(2x+3) = -\frac{1}{2} \cos(2x+3) + C$$

$dx = \frac{1}{2} d(2x+3)$

$$N8. \int (2x+3)^{20} dx = \frac{1}{2} \int (2x+3)^{20} d(2x+3) = \frac{1}{2} \frac{(2x+3)^{21}}{21} + C$$

$$N9. \int \cos(\alpha x + \beta) \cdot \cos(\alpha x - \beta) dx =$$

$$= \begin{cases} \cos \alpha \cdot \cos \beta \quad \text{①} \\ \cos(\alpha + \beta) = \cos \alpha \cos \beta - \sin \alpha \sin \beta \\ \cos(\alpha - \beta) = \cos \alpha \cos \beta + \sin \alpha \sin \beta \quad \text{②} \end{cases}$$

$$\cos \alpha \cdot \cos \beta = \frac{1}{2} (\cos(\alpha + \beta) + \cos(\alpha - \beta))$$

$$\text{①} \frac{1}{2} \int (\cos 2\alpha x + \cos 2\beta) dx = \text{const}$$

$$= \frac{1}{2} \int \cos 2\alpha x dx + \frac{1}{2} \int \cos 2\beta dx =$$

$$= \frac{1}{4\alpha} \sin 2\alpha x + \frac{1}{2} \cos 2\beta + C.$$

$$N10. \int \frac{dx}{2x^2 - 5x + 7} = \frac{1}{2} \int \frac{dx}{x^2 - \frac{5}{2}x + \frac{7}{2}} =$$

$$= \frac{1}{2} \int \frac{dx}{(x - \frac{5}{4})^2 - \frac{25}{16} + \frac{7}{2}} = \frac{1}{2} \int \frac{d(x - \frac{5}{4})}{(x - \frac{5}{4})^2 + \frac{31}{16}} =$$

$$= \frac{1}{2} \cdot \frac{4}{\sqrt{31}} \arctg \frac{(x - 5/4) \cdot 4}{\sqrt{31}} + C.$$

$$N11. \int \frac{dx}{\sqrt{x^2 - 2x + 5}} = \int \frac{dx}{\sqrt{(x-1)^2 - 1 + 5}} =$$

$$= \int \frac{d(x-1)}{\sqrt{(x-1)^2 + 4}} = \ln |(x-1) + \sqrt{x^2 - 2x + 5}| + C$$

$$= d\varphi(x)$$

$$N12. \int (\varphi(x))^\alpha \cdot \boxed{\varphi'(x) dx} =$$

$$= \int (\varphi(x))^\alpha d\varphi(x) = \int t^\alpha dt = \begin{cases} \frac{(\varphi(x))^{\alpha+1}}{\alpha+1} + C, & \alpha \neq -1 \\ \ln |\varphi(x)| + C, & \alpha = -1 \end{cases}$$

$$N13. \int \frac{6x - 7}{3x^2 - 7x + 1} dx = \int \frac{d(3x^2 - 7x + 1)}{3x^2 - 7x + 1} = \ln |3x^2 - 7x + 1| + C$$

$$\int \frac{f'(x) dx}{f(x)} = \ln |f(x)| + C$$

$$N14. \int \frac{3x - 2}{2 - 3x + 5x^2} dx =$$

$$= \int \frac{\frac{3}{10}(10x - 3) + \frac{9}{10} - 2}{2 - 3x + 5x^2} dx = \frac{3}{10} \int \frac{10x - 3}{2 - 3x + 5x^2} dx -$$

$$- \frac{11}{10} \int \frac{dx}{2 - 3x + 5x^2} = \frac{3}{10} \ln |2 - 3x + 5x^2| -$$

$$- \frac{11}{10 \cdot 5} \int \frac{dx}{x^2 - \frac{3}{5}x + \frac{2}{5}} = \frac{3}{10} \ln |2 - 3x + 5x^2| -$$

$$- \frac{11}{50} \int \frac{d(x - \frac{3}{10})}{(x - \frac{3}{10})^2 - \frac{9}{100} + \frac{2}{5}} = \frac{3}{10} \ln |2 - 3x + 5x^2| -$$

$$- \frac{11}{50} \cdot \frac{10}{\sqrt{31}} \arctan \frac{(x - \frac{3}{10})_{10}}{\sqrt{31}} + C$$

$$N15. \int \frac{dx}{ax^2 + bx + c} \quad - \text{non. reb.}$$

$$\int \frac{Ax + B}{ax^2 + bx + c} dx \quad \int \frac{Ax + B}{\sqrt{ax^2 + bx + c}} dx$$

$$\int \frac{P_n(x)}{Q_m(x)} dx$$

$$N15 \quad \int \frac{3x-6}{\sqrt{x^2-4x+5}} dx = \int \frac{\frac{3}{2}(2x-4)+6-6}{\sqrt{x^2-4x+5}} dx$$

$$= \frac{3}{2} \int \frac{2x-4}{\sqrt{x^2-4x+5}} dx = \frac{3}{2} \cdot 2 \sqrt{x^2-4x+5} + C.$$

$$\int \frac{f'(x) dx}{\sqrt{f(x)}} = \int \frac{d(f(x))}{\sqrt{f(x)}} = \int \frac{dt}{\sqrt{t}} = 2\sqrt{t} + C$$

$$= 2\sqrt{f(x)} + C$$

$$N16. \quad \int \frac{\textcircled{x} dx}{(1-x^2)^2} = \frac{1}{2} \int \frac{d(x^2)}{(1-x^2)^2} = \frac{1}{2} \int \frac{d(-t+1)}{(1-t)^2} =$$

$$= -\frac{1}{2} \int \frac{dz}{z^2} = +\frac{1}{2} \frac{1}{z} + C =$$

$$= \frac{1}{2} \cdot \frac{1}{1-x^2} + C.$$

$$x \cdot dx = \frac{1}{2} d(x^2)$$

$$N17. \quad \int \left(\frac{x}{x^5+2} \right)^4 dx = \int \frac{x^4}{(x^5+2)^4} dx =$$

$$= \frac{1}{5} \int \frac{d(x^5+2)}{(x^5+2)^4} = \frac{1}{5} \int \frac{dt}{t^4} = -\frac{1}{3 \cdot 5} t^{-3} + C =$$

$$= -\frac{1}{15 (x^5+2)^3} + C.$$